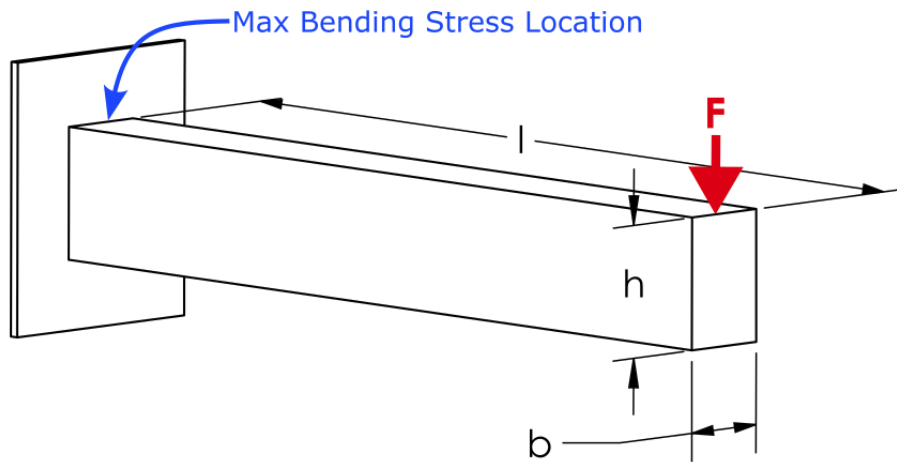


Database Consistency Test

Introduction

This EngineeringPaper.xyz sheet shows how to calculate the maximum stress and displacement for a cantilever beam loaded by a vertical force at its end. The geometry of the beam is shown in the figure below. The beam has a rectangular cross section with height h and width b . The maximum stress in a cantilever with length much larger than its height is due to bending stress that occurs at the base of the beam. For a downward force, the maximum stress will be a tensile force at the top of the beam and a compressive stress at the bottom of the beam. For beam sections that are symmetric top to bottom, the maximum tensile and compressive stresses will have equal magnitude. #



Select Beam Material

Select Beam Section

Source for beam section equations and images: https://en.wikipedia.org/wiki/List_of_second_moments_of_area

Define the Other Beam Parameters

$$l = 500 [mm]$$

$$F = 500 [N]$$

Define the Deflection and Stress Equations

The maximum displacement of a cantilever beam loaded at its end is defined by [1]:

$$y_{max} = -\frac{F \cdot l^3}{3 \cdot E \cdot I_x}$$

where I_x is the area moment of inertia of the cross section about the x-axis as selected in the table above. The maximum displacement can now be queried:

$$y_{max} [mm] = -0.165297794307557 [mm]$$

The maximum bending stress in a beam section is given by:

$$\sigma_{max} = \frac{M \cdot c}{I_x}$$

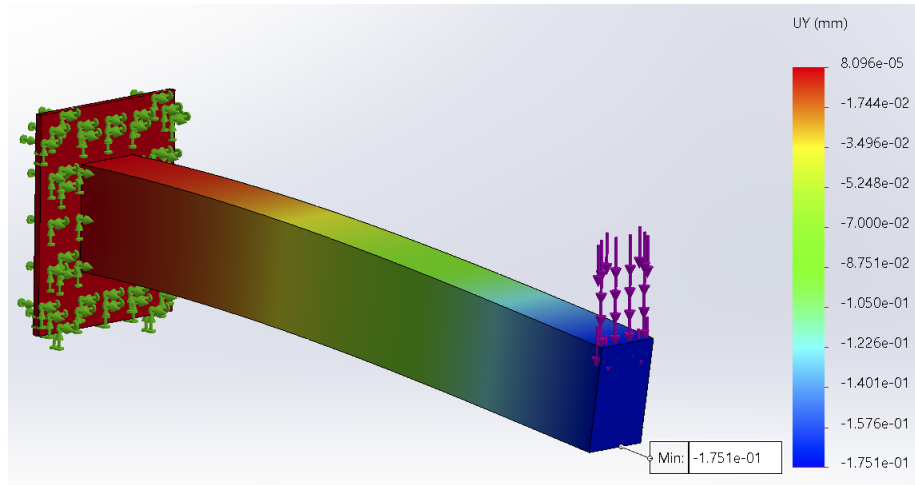
where M is the bending moment at the section of interest and c is distance from the neutral axis of the beam to the top or bottom of the beam. The maximum moment occurs at the base of the beam and is defined by:

$$M = l \cdot F$$

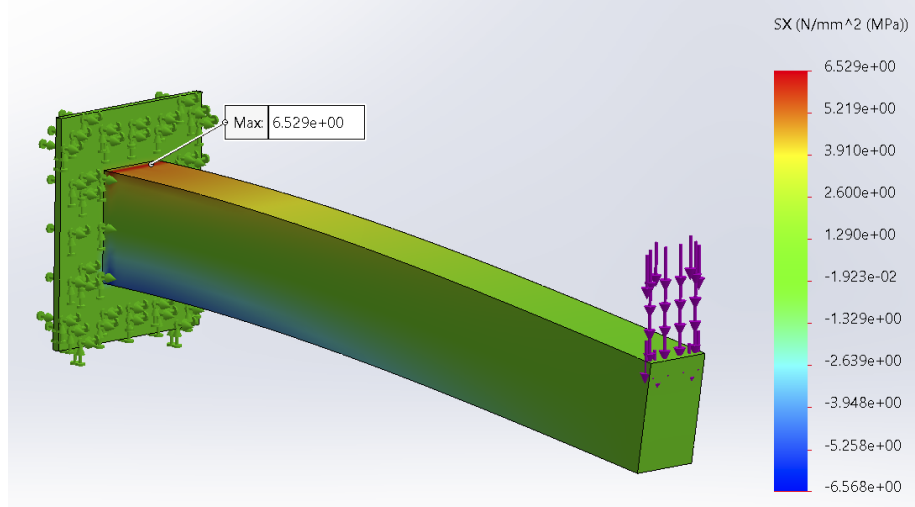
where l is the length of the beam. Now that everything is defined, the maximum stress can be queried and converted to [MPa] units:

$$\sigma_{max} [MPa] = 5.33333333333333 [MPa]$$

These results can be compared to the results obtained from a finite element analysis using SolidWorks. The beam has the same dimensions, loading, and material as the rectangular cross-section and the aluminum table options above. The displacement plot is shown below. The maximum displacement magnitude obtained using finite element analysis is .175 [mm], which is close to the .165 [mm] obtained using the equations above.



Similarly, the maximum stress calculation can be compared to the SolidWorks results. The maximum stress obtained from SolidWorks is 6.5 [MPa] as shown below. Note that this is higher than the 5.3 [MPa] obtained using the beam equation. This is the result of the stress concentration that occurs at the base of the cantilever where it meets the fixed boundary condition. A little further from the fixed boundary condition, the finite element values are close to the calculated values.



[1] Norton, R. L. "Machine design. A integrated approach, 5th Editi." (2013).

$$y = \frac{F}{6 \cdot E \cdot I_x} \cdot (x^3 - 3 \cdot l \cdot x^2)$$